Chemistry of Dyes

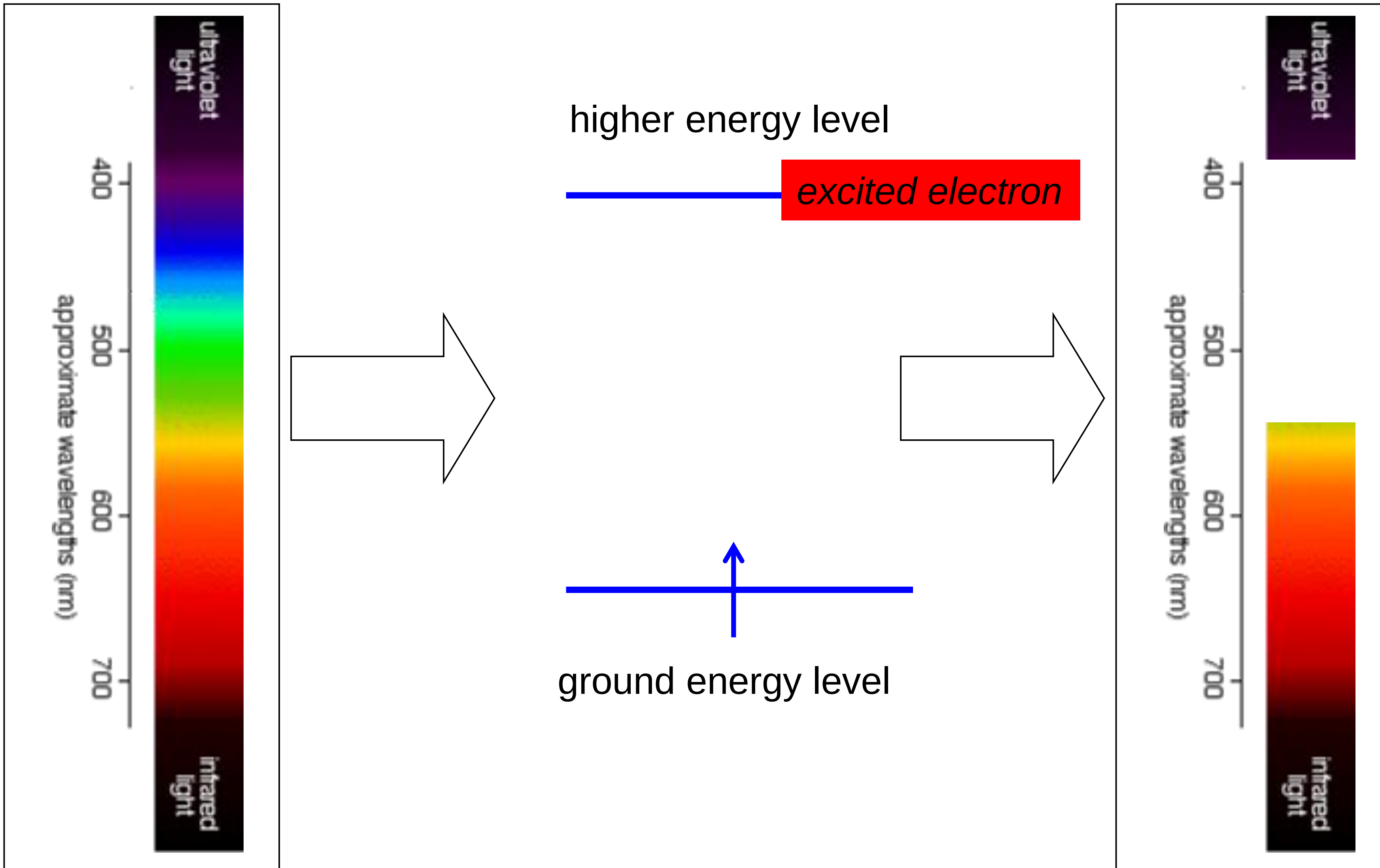
Dyes

- Dyes are usually colored organic compounds that may be used to impart color to a substrate like paper, cloth, leather or plastic
- Imparted colors are reasonably washfast and lightfast
- contain chromophores: parts of the molecule that absorb light in the visible spectrum
- Chromophores occur in organic molecules when conjugated systems allow for delocalised electrons

Chromophores

- either C=C or N=N or C=O groups next to each other in a molecule
- the electrons in these double bonds spread out across them all – they are delocalised

Light energy is absorbed by excited electrons



Types of Dyes

- the main way dyes are classified is by the method that they are applied to the fabric
- the types of dye are:
  - acid dye
  - direct dye
  - reactive dye
  - vat dye
  - disperse dye
  - basic dye

- for each of these types of dye, a whole range of colours is possible

Colour fastness

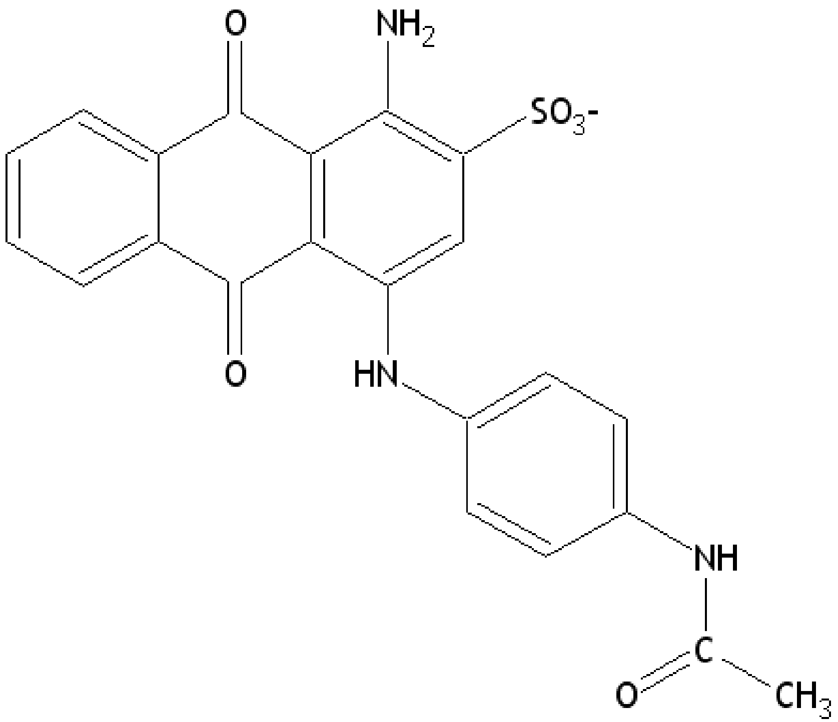
- it is important that a dye does not wash out of a fabric – it is colour fast
- the dye must also be chemically stable and not affected by light
- the colour fastness depends on:
  - the forces of attraction between the dye molecules and the fabric molecules

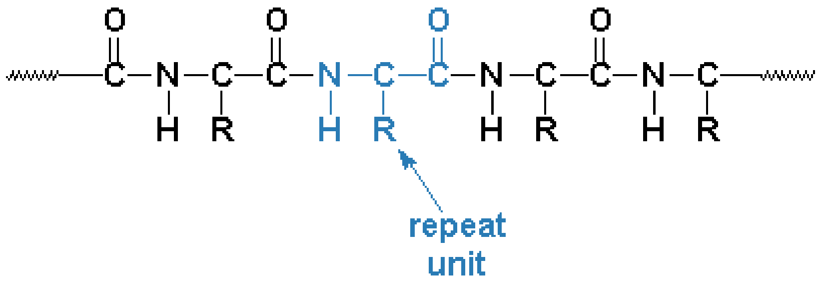
How dyes bond to fabrics:
1. acid dyes

- usually have a $–SO_3^-$ or $–COO^-$ group
- they bond well to protein fibres such as wool or silk
- they don't bond well to cellulose fibres
- the bonding is due to the attraction between the negatively-charged groups on the dye and positively charged groups on the fibre

…..acid dyes

- examples of an acid dye and a protein fibre:





•silk or wool fibre (-R is a different group depending on which fibre)

•in acid conditions, the –N-H groups become charged: $-NH_2^+$

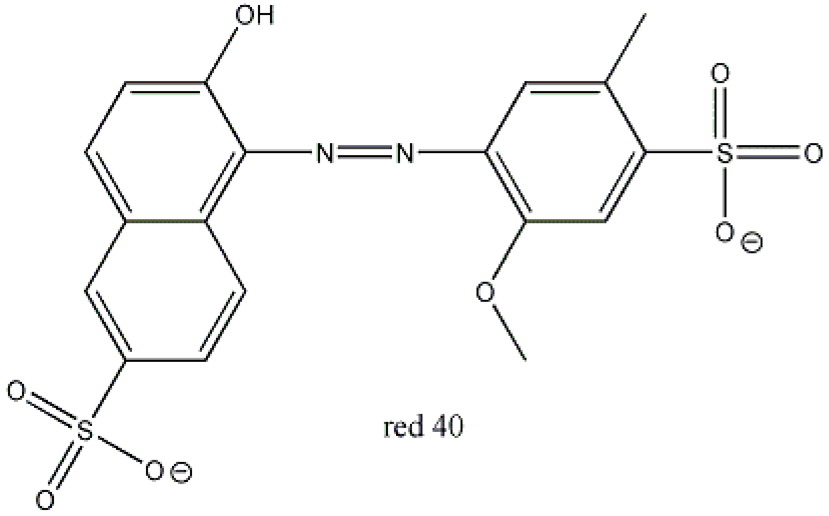
•nylon is a man-made version of a protein fibre

2. direct dyes

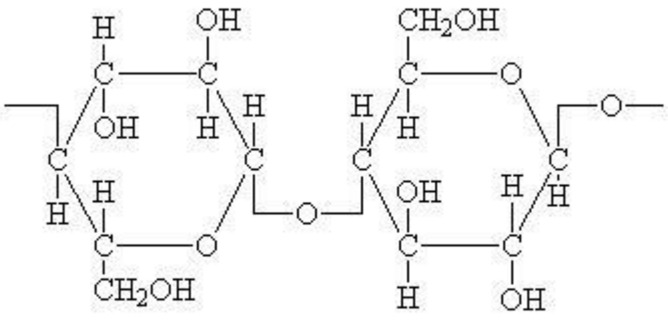
- get their name as they were the first dyes to be put directly onto fabrics
- they are used on cellulose fibres (main one is **cotton**)
- usually azo-dyes (this means they contain a –N=N- group)
- the bonding is by weaker intermolecular forces and hydrogen bonds
- direct dyes sometimes aren’t that colourfast

……direct dyes

- examples of a direct dye and a cotton fibre:



red 40



Cellulose

•cotton fibre (cotton is cellulose)

3. reactive dyes

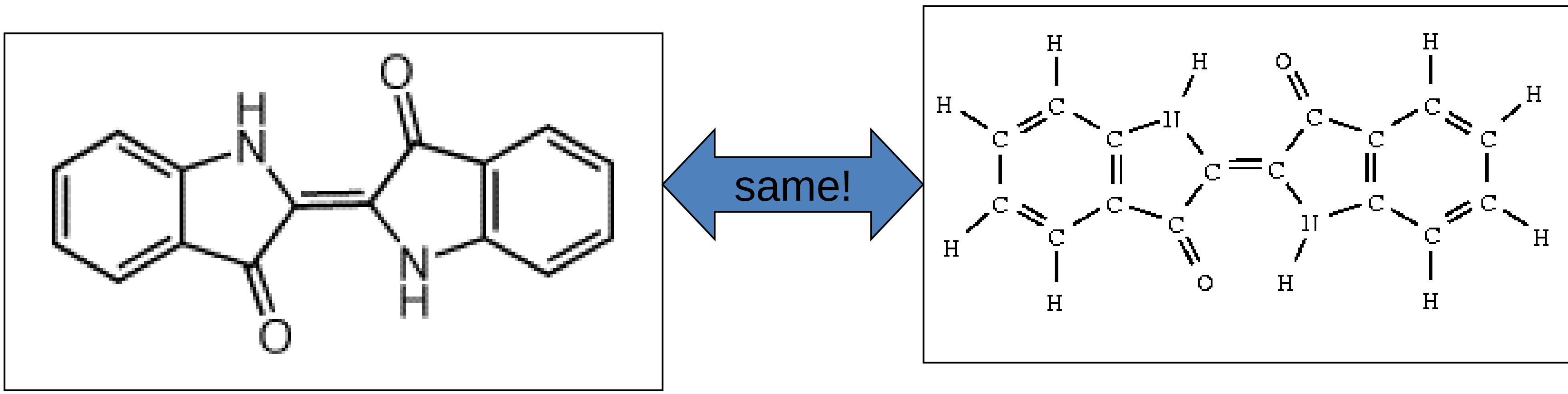
- they react with the fabric to form a chemical (covalent) bond with it
- this strong bond gives outstanding colour-fastness
- they are a modern development in dyes
- different dyes have been designed to react with different fabric molecules

4. vat dyes

- Vat dyes don’t dissolve in water in their normal form
- a chemical reaction is carried out (reduction) to make them water-soluble
- they are then added to the fabric
- the chemical reaction is reversed so they dye returns to its original form (and colour)

.....vat dyes

- example of a vat dye:



•indigo is reduced before it can be added to fabrics, and turns yellow

•it then oxidises again in air and turns back to blue

•vat dyes molecules often have lots of ring systems and not many functional groups

•the chemical reaction may change functional groups irreversibly

•bonding is by inter-molecular forces

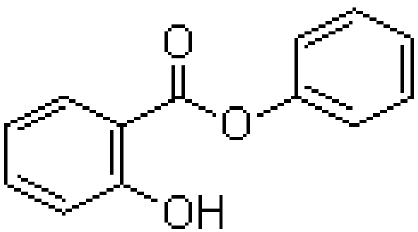
•often used with cellulose (cotton)

5. disperse dyes

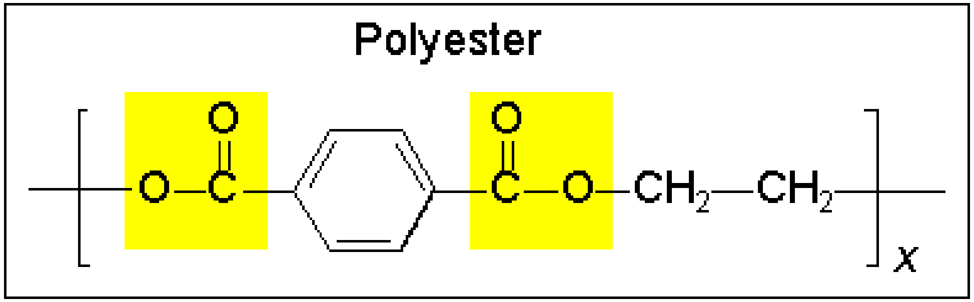
- Disperse dyes are usually small molecules don’t dissolve well in water
- they don’t have charged groups on them
- they bond to polyester well because they can get in between the tightly-packed polymer chains

…. disperse dyes

- example of a disperse dye and polyester:



•disperse dye ‘blue’



•polyester molecules always have the ester groups, shown in yellow

•a man-made fibre

•polar groups on the disperse dye are attracted to polar groups on the polyester

6. basic dyes

- Basic dyes have positively-charged (cationic) groups on them
- this means they are attracted to fabric molecules with negative charges
- the main fabric in this case is acrylic

…. basic dyes

- example of a basic dye:

•mauveine b, the first synthetic dye is a basic dye (look for the positive charge)